

- 3 (a) (i)** $E_K = \frac{1}{2}mv^2$
- $$= 0.5 \times 0.40 \times 0.30^2$$
- $$= 1.8 \times 10^{-2} \text{ J} \quad \textbf{A1}$$
- (ii)** (change in) kinetic energy = work done on spring / (change in) elastic potential energy **C1**
- $$1.8 \times 10^{-2} = \frac{1}{2} \times F \times 0.080 \quad \textbf{C1}$$
- $$F_{\text{MAX}} = 0.45 \text{ N} \quad \textbf{A1}$$
- (iii)** $a = F / m = 0.45 / 0.40$
- $$= 1.1 \text{ m s}^{-2} \quad \textbf{A1}$$
- (iv)** 1. constant velocity / resultant force is zero, so in equilibrium **B1**
2. decelerating / resultant force is not zero, so not in equilibrium **B1**
- (b)** curved line from the origin **M1**
- with decreasing gradient **A1**

